Some Thoughts on Mathematical Understanding

Looking back on my years in graduate school, I realize how much I learned from the different research styles and personalities of my professors across different areas of maths. From my algebra professors, I learned the importance of working carefully through details, understanding things, finding simple and elegant proofs, and developing a particular perspective on the subject. From my number theory professors, I learned to take a broad and open-minded approach to different ideas and methods, and, even more, to find and formulate questions, and to have the patience to start with small questions and gradually build them up.

In commutative algebra and algebraic K-theory, one thing I have always appreciated about my algebra professors is seeing how in practice the simplest proofs can sometimes reflect the deepest understanding. A proof is not only a certificate that a statement is true; the way it is organized can reveal how one sees the underlying structures. This is part of what makes commutative algebra so beautiful to me. Its language can be remarkably simple, but there are hidden structures and rich mathematical phenomena beneath it. There is something deeply aesthetic about the subject to me. And I don't feel that the way I approach and enjoy maths will change very much despite the development of AI.

Personally, I prefer not to use AI when doing maths research, although I find it quite useful in many other aspects of life as it helps with many practical problems. When it comes to maths, the process feels different to me. More generally, when it comes to things I have a strong personal sense of, I prefer not to involve AI in a substantial way. That said, I don't think collaborating with AI is a bad thing. It is simply a different research style, which can be genuinely fruitful and stimulate new ways of thinking, depending on personal preferences and perhaps also on the particular area of maths one works in.

I also don't think that mathematicians who choose not to use AI will necessarily be at a disadvantage in their research or find that all their problems have already been solved by AI. There are always plenty of questions to ask and investigate beyond a stated conjecture, many of which have not even been formulated explicitly or made known to others. Sometimes seeing an interesting problem and exploring where it leads is itself the beginning of a new development. In this sense, I don't think there is any need to feel that one is racing against AI.

I think the development of AI offers new ways of doing research for everyone to try and experiment with, and ultimately to decide which approaches suit them best. I don't think any one research style will necessarily be at a disadvantage compared with others. It is largely a matter of individual preferences and ways of thinking.

AI in solving conjectures

Recently, there have been many AI-generated attempts at major mathematical conjectures. This is an interesting development, although we don't yet know whether all these arguments are correct, and I think many of them still need serious verification. But regardless of whether these conjectures are solved, this does not mean that the mathematical directions surrounding them are closed off. There are always many more questions to explore and build on.

I think what makes a conjecture interesting is often the possibility of exploring new mathematical territory, either along the way to a solution or through what the solution itself might reveal. A conjecture usually points toward some underlying structure or theory that we are trying to understand. Sometimes a proof reveals that structure clearly, while other times it settles the conjecture without explaining nearly as much. A counterexample can leave even more

questions behind, since it often tells us that something fails without explaining why or under what conditions the conjecture might still hold.

AI-generated maths can advance mathematical knowledge, but even when AI generates a proof, further mathematical understanding and insight are often needed to verify it, see what it reveals, and explore how the ideas can be developed further.

Mathematical Insights

Mathematical research involves more than proving difficult statements. Sometimes a mathematician looks at a theorem and feels that it has been formulated in the wrong way. A proof may be correct but still fail to explain the phenomenon.

I often heard my professors say things like, "There should be a more conceptual explanation," or "There should be a much simpler proof from the right perspective." Such judgments are difficult to formalize. They are not simply reactions to a proof being too long or the hypotheses too complicated; they arise from an internal picture of the subject built through mathematical understanding and experience.

That internal picture develops slowly and individually. Over time, we develop a personal sense of how a subject behaves, shaped by the problems we have worked on, the mistakes we have made, and the different perspectives we have encountered. Mathematical intuition is not just a collection of facts, but also an individual way of organizing them, recognizing what matters, and sensing the structures behind them. There is something profoundly human and individual in this process.

AI's ability to draw on a large amount of existing literature certainly has its advantages, but it does not necessarily lead to the most insightful approach. Of course, things can be quite different when mathematicians bring their own understanding and perspectives to AI-generated ideas, refining them or developing them in new directions.

Collaboration with AI

Maths has always developed through different styles of collaboration. Now it may develop through a new form of collaboration with AI, but how mathematicians choose to work with AI will also depend on individual preferences. Just as some prefer frequent discussion and collaboration while others work mostly alone, some may find constant interaction with AI productive, while others may use it only occasionally or not at all. Whatever style one chooses, mathematical understanding will remain important, including for judging, modifying, and developing what AI produces.

But I have never believed that maths would not matter simply because AI can do certain things better than humans, or that mathematicians who choose not to use AI would be at a disadvantage. Maths is after all a deeply communal activity. Much of its joy comes not only from solving difficult problems, but also from understanding ideas, appreciating their beauty, and communicating them to others. Different people may find enjoyment in different aspects of maths, but I do not think our motivation should only depend on the pride of being uniquely capable of doing something. Mathematical knowledge and understanding can remain interesting regardless of whether they come from a human or assisted by AI.

Of course, AI raises real questions about credit, misuse, education, and responsibility, but much also depends on how people choose to use it. Mathematicians can genuinely engage with AI and produce substantial work through interactive collaboration. In such cases, there may not

be, and perhaps should not be, a sharp division between human and AI contributions. After all, even in traditional mathematical collaborations, credit is not simply assigned according to measurable contributions, and the intellectual contributions of different collaborators are often deeply intertwined.

On the other hand, there may also be cases where AI makes the predominant contribution. I think this is a rather different situation, but such work can still be appreciated if the human authors have assimilated the ideas and developed a comparable understanding. Mathematical understanding remains essential, and I don't think it is sustainable or possible, either for individual researchers or for the mathematical community, to accept predominantly AI-generated work without understanding.

Whatever forms the mathematical community may find to adapt to these changes, I think the values that make maths meaningful will remain, and may even be deepened as we reconsider the relationship between humans and AI. I genuinely don't think we need to be overly concerned about the situation, and I hope that everyone can continue to enjoy maths just as much, regardless of how AI develops. We can continue to enjoy the maths we care about, deepen our understanding, share ideas, and follow the questions that excite us. These are the things that have always mattered, and I believe they will continue to matter, both to our heart and to the mathematical community.

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